Lab 4: PageRank Algorithm

Information Retrieval (417524B) - SPPU BE AI&DS

Implement Page Rank Algorithm. (Use Python or Beautiful Soup for implementation).

CODE

```
from bs4 import BeautifulSoup

pages = {
    "A": '<a href="B">B</a><a href="C">C</a>',
    "B": '<a href="C">C</a>',
    "C": '<a href="A">A</a>',
    "D": '<a href="A">A</a><a href="C">C</a>',
}

links = {}
for name, html in pages.items():
    soup = BeautifulSoup(html, "html.parser")
    links[name] = [a["href"] for a in soup.find_all("a")]

print("Link graph:")
for k, v in links.items():
    print(f"  {k} -> {v}")

nodes = list(pages)
N = len(nodes)
d = 0.85
rank = {n: 1.0 / N for n in nodes}

for it in range(1, 101):
    new = {n: (1 - d) / N for n in nodes}
    for n in nodes:
        if links[n]:
            share = d * rank[n] / len(links[n])
            for t in links[n]:
                new[t] += share
        else:
            for t in nodes:
                new[t] += d * rank[n] / N
    diff = sum(abs(new[n] - rank[n]) for n in nodes)
    rank = new
    if diff < 1e-6:
        break

print(f"\nConverged in {it} iterations")
print("PageRank scores:")
for n, s in sorted(rank.items(), key=lambda x: -x[1]):
    print(f"  {n}: {s:.4f}")

# d = 0.75
```

OUTPUT

```
Link graph:
  A -> ['B', 'C']
  B -> ['C']
  C -> ['A']
  D -> ['A', 'C']

Converged in 28 iterations
PageRank scores:
  C: 0.3839
  A: 0.3797
  B: 0.1989
  D: 0.0375
```